This report investigates income classification on the Adult Income dataset using mixed Naïve Bayes. The task is to predict whether an individual earns more than $50K per year from demographic and employment features. First, a supervised model is trained and analysed to examine what it learns from the labelled data. It is then extended using semi-supervised label propagation with unlabelled data. Finally, the supervised and semi-supervised models are compared in terms of performance, confidence, and learned parameters.

Q1

The supervised training data shows clear class imbalance. The prior probability of *<=50K* is 0.7541, while the prior probability of *>50K* is 0.2459. As shown in Figure 1, the low-income class is much larger than the high-income class. This means the model is trained on many more *<=50K* examples, so it is likely to be biased toward the majority class. Since Naïve Bayes uses class priors directly in posterior estimation, this imbalance may affect later predictions.

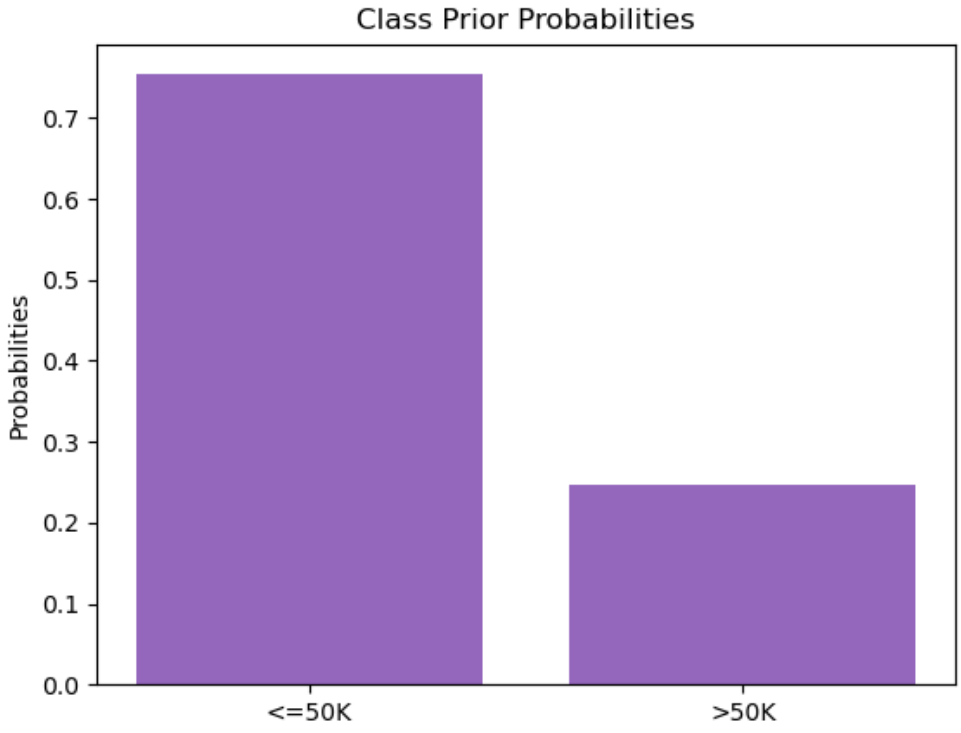
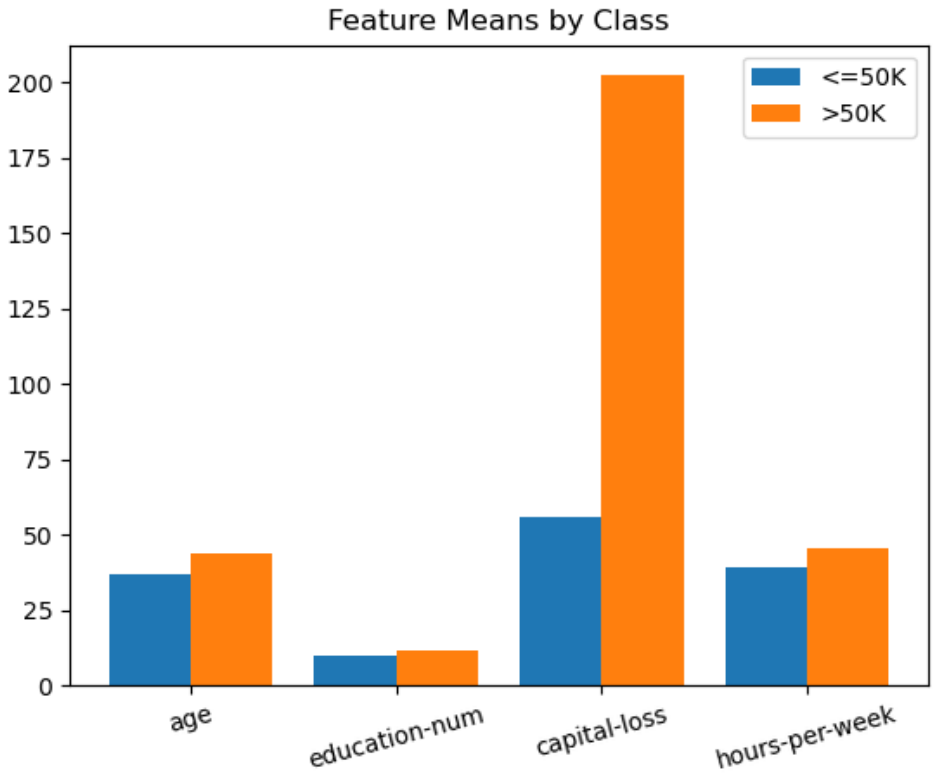


Figure 1 Class Distribution in Supervised Training Set

For the continuous features, the >50K class has higher mean values for all five variables. As shown in Figure 2, *education-num*, *age*, and *hours-per-week* separate the two classes clearly, with the high-income class having larger means in each case. *capital-gain* shows the strongest numerical difference, although its very large

standard deviation in Table 1 suggests a highly skewed distribution. These patterns are intuitive, since higher education, older age, and longer working hours are all plausibly associated with higher income.

Figure 2 Mean values of key continuous features for each income class.



Feature	Mean(<=50K)	Std(<=50K)	Mean(>50K)	Std(>50K)
Age	37.05	13.71	43.94	10.30
Education-num	9.63	2.45	11.59	2.36
Capital-gain	157.67	1017.86	3607.15	13616.62
Capital-loss	55.97	316.02	202.36	603.99
Hours-per-week	39.43	11.91	45.64	10.40

Table 1 Estimated Gaussian parameters for the continuous features in supervised model

For the categorical features, Table 2 shows that the strongest predictors of *>50K* include *Prof-school*, *Doctorate*, *Masters*, *Self-emp-inc*, and *Taiwan*, all with large R values. In contrast, the strongest predictors of *<=50K* include *Own-child*, *Priv-house-serv*, *Other-relative*, *Other-service*, and *9th*, all with very small R values. Overall, the learned parameters appear sensible, and education-related features seem especially

useful for classification

Class	Feature	Value	R
>50K	education	Prof-school	11.501389
	education	Doctorate	10.578395
	education	Masters	4.852067
	workclass	Self-emp-inc	4.460052
	native-country	Taiwan	4.437463
<= 50K	relationship	Own-child	0.039766
	occupation	Priv-house-serv	0.062934
	relationship	Other-relative	0.102446
	occupation	Other-service	0.102832
	education	9th	0.104989

Table 2 Top five predictive categorical values for each class based on the posterior ratio R

Q2

The supervised mixed Naïve Bayes model achieved an overall accuracy of 0.83 on the test set. As shown in Figure 3, the confusion matrix indicates that the model performs much better on the *<=50K* class than on the *>50K* class. In particular, 11729 low-income instances were classified correctly, while 2212 high-income instances were misclassified as *<=50K*. This suggests that the classifier tends to favour the majority class, which is consistent with the class imbalance observed in Q1.

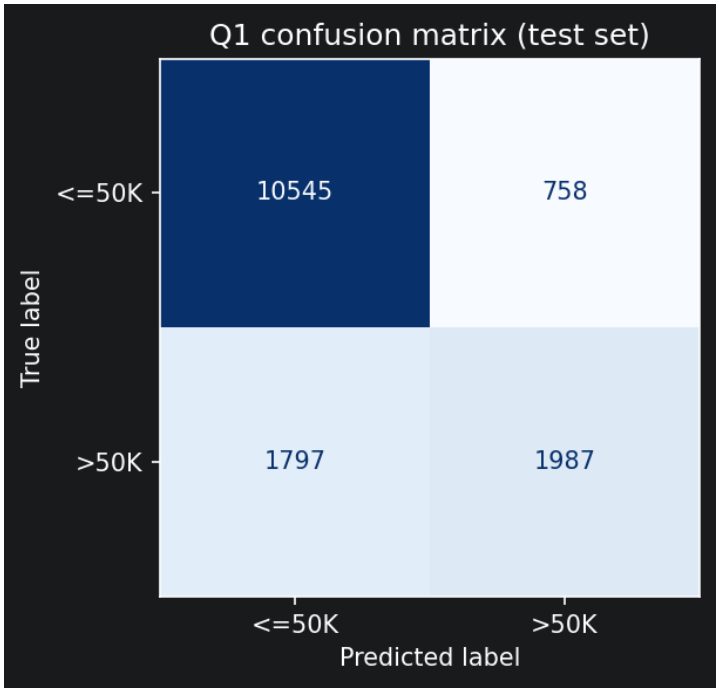


Figure 3 Confusion Matrix of the supervised model on the test set.

The class-level metrics show the same pattern. As reported in Table 3, the model performs strongly on *<=50K* but recall for *>50K* is much lower. This means the classifier is reliable for the majority class, but it misses many truly high-income cases. Therefore, *>50K* is clearly the harder class to classify correctly.

	Precision	Recall	F1-score	Support
<= 50K	0.84	0.95	0.89	12337
>50K	0.74	0.44	0.55	3944
Accuracy			0.83	16281

Table 3 Classification Report of the Supervised Model on the test set.

There are 1195 out of 16281 test instances containing at least one unseen categorical value (OOV). These instances were still classified, so no test cases were skipped entirely. Representative confidence examples are summarised in Table 4. High-confidence *>50K* predictions usually include strong indicators such as *Doctorate* or *Prof-school*, long working hours, and extremely large capital-gain values. In contrast, high-confidence *<=50K* examples are much younger, have lower education levels, and usually have zero capital gain. Cases near the decision boundary contain mixed signals, so the model is much less certain.

Group	Log_R	True label	Predicted label	Key features(age, education, occupation, marital status, hours-per-week, capital-gain)
High-confidence > 50K	4789	>50K	>50K	41, Doctorate, Prof-specialty, Married-civ-spouse, 70, 99999
	4789	>50K	>50K	50, Prof-school, Prof-specialty, Married-civ-spouse, 55, 99999
	4789	>50K	>50K	50, Prof-school, Prof-specialty, Married-civ-spouse, 80, 99999
High-confidence <= 50K	-21.92	<=50K	<=50K	18,10th, Other-service, Never-married, 3, 0
	-21.77	<=50K	<=50K	17, 10th, Other-service, Never-married, 12, 0
	-21.77	<=50K	<=50K	17, 11th, Other-service, Never-married, 5, 0
Near boundary	1.001	<= 50K	>50K	41, Masters, Prof-specialty, Married-civ-spouse, 50, 0
	1.002	>50K	>50K	50, Masters, Exec-managerial, Married-civ-spouse, 50, 0
	1.002	>50K	>50K	50, Masters, Exec-managerial, Married-civ-spouse, 50, 0

Table 4 Examples of high-confidence and near-boundary predictions from the supervised model

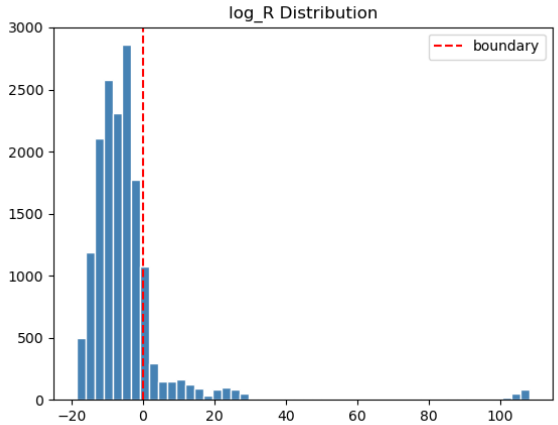


Figure 4 Distribution of the log posterior ratio on the test set for the supervised model.

Q3

For the semi-supervised extension, Option 1 (Label Propagation) was selected. This approach is consistent with the supervised mixed Naïve Bayes model from Q1, insofar as the model already produces posterior probabilities that can be used to assign pseudo-labels to unlabelled instances. Initially, the supervised model was trained on the labelled training data. Subsequently, it was employed to predict class labels for the unlabelled dataset. However, it was not assumed that all pseudo-labels were equally reliable. Conversely, the absolute value of the log posterior ratio was utilised as a confidence measure, ensuring that only the most confident pseudo-labelled instances were incorporated into the training set. This was intended to reduce the risk of reinforcing incorrect predictions from the original supervised model.

In order to select the confidence threshold, the approach adopted was to follow the assignment requirement and utilise a validation set as opposed to tuning directly on the test set. An evaluation was conducted in which a range of thresholds were compared, and their validation F1-scores were calculated. As demonstrated in Figure 5, there was an enhancement in performance when very low-confidence pseudo-labels were excluded, which suggests that noisy pseudo-labels can be detrimental to the semi-supervised model. The optimal threshold was determined to be 12, which yielded a validation F1-score of 0.5803. Reducing the thresholds resulted in the addition of a greater number of pseudo-labelled instances; however, this also led to an increase in noise due to the inclusion of numerous uncertain predictions. Conversely, the application of exceedingly elevated thresholds yielded a substantially diminished pseudo-labelled set, thereby diminishing the efficacy of utilising unlabelled data. This demonstrates a clear trade-off between the quantity of additional training examples and the quality of their labels.

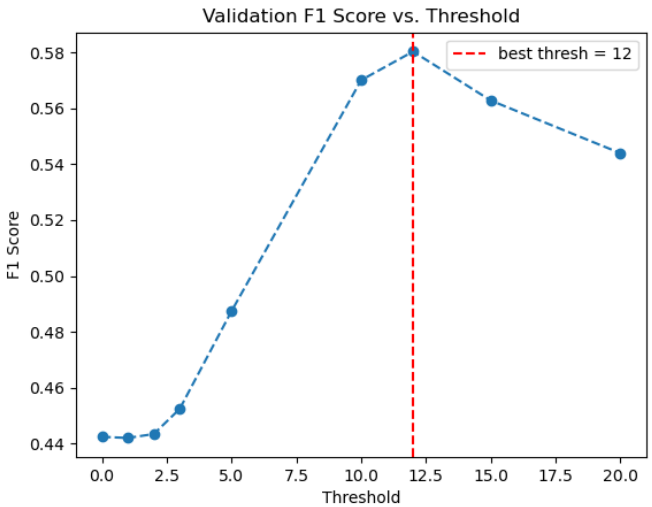


Figure 5 Validation F1 score under different confidence threshold

Utilising the designated threshold, the final semi-supervised training set comprised 18,841 instances, encompassing 3,765 pseudo-labelled examples. As demonstrated in Figure 6, the majority of selected pseudo-labels were classified as ≤50K, thereby reflecting the pronounced class imbalance that was already evident in both the labelled data and the predictions of the supervised model. This phenomenon can be attributed to the fact that a model that has been trained on imbalanced data is more likely to assign the majority class more frequently. Nevertheless, the thresholding process ensured that only the most confident pseudo-labels were used. It is evident that, in general, the implementation is uncomplicated, exhibits high computational efficiency, and is robustly substantiated by the validation results. However, it is important to note a significant limitation of this approach: pseudo-labels do not constitute true labels. Consequently, any bias or systematic error present in the supervised model may be partly propagated into the semi-supervised model. Nevertheless, the threshold-based selection step has been demonstrated to assist in the management of this risk, thereby enhancing the final approach's robustness.

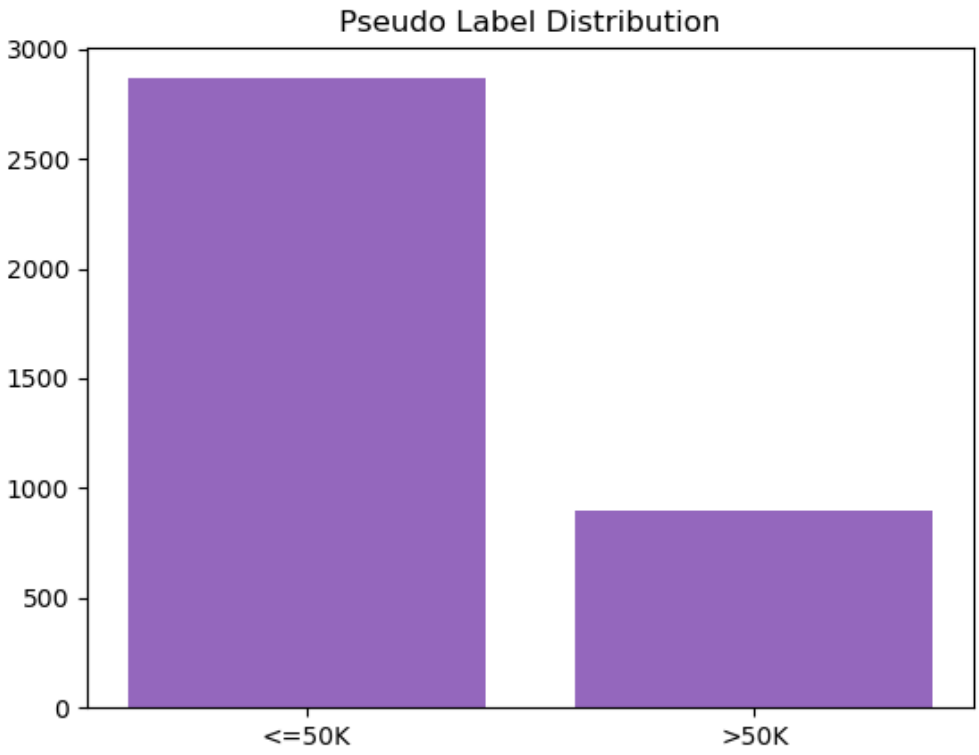


Figure 6 Class distribution of selected pseudo-labelled instances

Q4

The semi-supervised model achieved the same overall accuracy as the supervised model (0.83), so its overall performance remained unchanged. However, the class-level results show a more meaningful difference. For the <=50K class, precision increased slightly from 0.84 to 0.85, while recall decreased slightly from 0.95 to 0.94, so the F1-score remained almost unchanged at 0.89. For the >50K class, precision decreased from 0.74 to 0.71, but recall increased from 0.44 to 0.49, and the F1-score improved from 0.55 to 0.58. This shows that the semi-supervised model identified more truly high-income individuals, even though it also introduced some additional false positives. Since the supervised model was clearly weaker on the minority class, this trade-off is useful rather than harmful. A summary of the main test-set metrics for both models is given in Table 5.

Metric	Supervised	Semi-supervised
Accuracy	0.83	0.83
Precision (<=50K)	0.84	0.85
Recall (<=50K)	0.95	0.94
F1-score (<=50K)	0.89	0.89
Precision (>50K)	0.74	0.71
Recall (>50K)	0.44	0.49
F1-score (>50K)	0.55	0.58
Macro average F1	0.72	0.73
Weighted average F1	0.81	0.82

Table 5 Performance comparison between the supervised and semi-supervised models on the test set.

This alteration is also evident in the confusion matrix. As demonstrated in Figure 7, the semi-supervised model yielded a greater number of true positives for >50K and a lower number of false negatives in comparison to the supervised model. In other words, the number of high-income cases that were missed was lower. This enhancement signifies a substantial advancement, as the primary deficiency of the supervised model in Q2 was its suboptimal recall for the high-income category. Despite the absence of an increase in overall accuracy, the semi-supervised model demonstrated an

enhancement in the balance of its treatment of the two classes. The findings thus indicate that the supplementary pseudo-labelled data predominantly facilitated the model's adjustment of its decision boundary, thereby enabling the recovery of a greater number of minority-class instances. This enhancement was anticipated to a certain degree, as the objective of label propagation is to leverage supplementary structure in the unlabelled data, particularly when the labelled data alone is restricted or imbalanced.

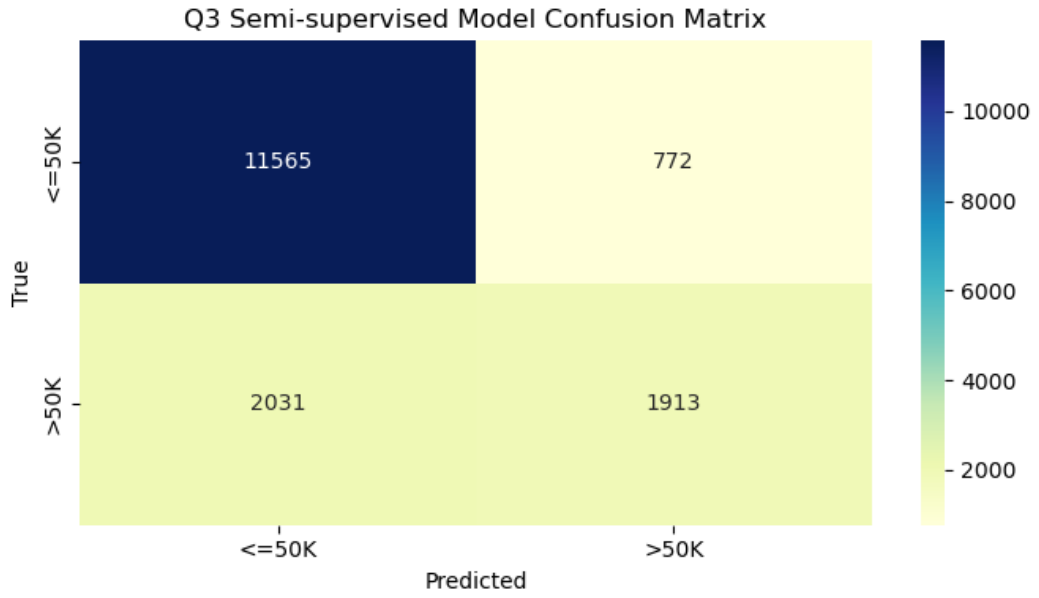


Figure 7 Confusion Matrix of the semi-supervised model on the test set.

The model parameters also changed after semi-supervised training. As shown in **Figure 8**, the mean values of several continuous features shifted, especially for capital-gain and capital-loss in the >50K class. This suggests that the semi-supervised model placed stronger emphasis on financial indicators when distinguishing high-income from low-income cases. The confidence distribution also changed, as shown in **Figure 9**, which indicates that the model's posterior structure was adjusted after semi-supervised learning. However, the overall pattern of the learned predictors remained broadly similar to the supervised case. The strongest education- and occupation-related indicators still remained important, so the semi-supervised model did not fundamentally change what the classifier had learned. Instead, it refined the original supervised model and improved its sensitivity to the harder >50K class. Overall, the improvement is modest rather than dramatic, but it is still meaningful because it improves minority-class performance without reducing overall accuracy.

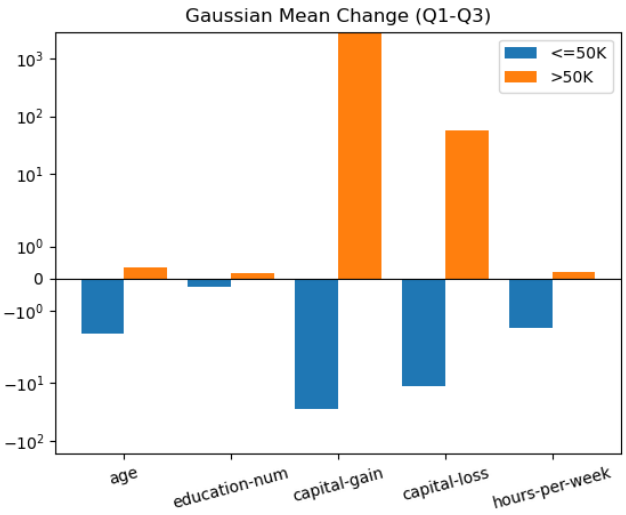


Figure 8 Change in Gaussian feature means from supervised to semi-supervised model (Q1 to Q3)

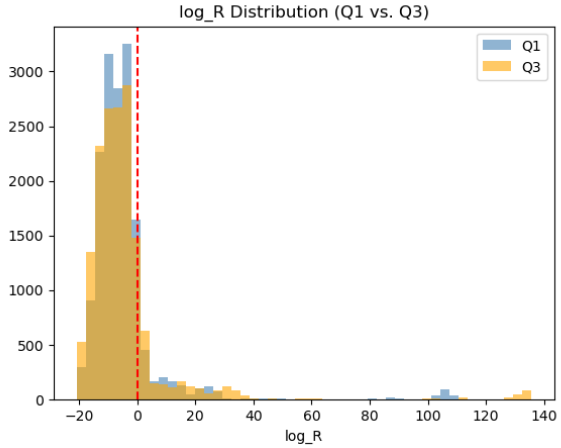


Figure 9 Distribution of log posterior ratios for the semi-supervised model

In conclusion, the mixed Naïve Bayes model demonstrated satisfactory overall performance on the income classification task, although it was evidently more robust in the majority ≤50K class than in the >50K class. The semi-supervised label propagation approach yielded a modest enhancement in minority-class performance, particularly in terms of recall and F1-score for >50K. The findings demonstrate that unlabelled data can be advantageous when incorporated with confidence-based pseudo-labelling.